# Who Moves the Price? Trading-Session Return Decomposition and Cross-Session Momentum in Gold and Silver Markets

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## Abstract

We decompose the cumulative returns of gold (XAU/USD) and silver (XAG/USD) across three global trading sessions, Asia (00:00–08:00 UTC), Europe (08:00–14:30 UTC), and the United States including the late session (14:30–24:00 UTC), using hourly spot prices from 22 July 2024 to 7 August 2026, and we backtest a session-to-session momentum rule on the same data. The sample is a strong one for this question: gold rose about 81% and silver about 119% over it. The Asian session produced 77.2% of gold's cumulative log return and 61.7% of silver's. Europe added a further positive share (+36.2% and +46.6%), while the US session, despite being the most volatile of the three, subtracted (−13.4% and −8.2%). Asia's contribution is positive in every calendar year, and the severe 2026 correction occurred almost entirely during US hours. Session returns show short-horizon momentum concentrated in continuation into the Asian session (lag-1 autocorrelation of +0.13 for gold, +0.12 for silver). Before costs, a session-momentum strategy beats buy-and-hold; after a 0.02% charge per position change, only the long-only variant survives, with a net Sharpe of 1.68 for gold at roughly half the buy-and-hold drawdown, and 0.84 for silver with the drawdown cut from −53.6% to −26.2%. Momentum run inside the US session loses money net of costs, which fits the picture of US hours as high-variance, drift-less information trading. Taken together, the results are consistent with marginal gold and silver pricing having shifted eastward in the post-2022 regime of central-bank accumulation.

**Keywords:** gold, silver, trading session, return decomposition, price discovery, intraday momentum, central bank gold demand

**JEL Classification:** G12, G14, G15, F31, Q02

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## 1. Introduction

Since 2022, the gold market has been reshaped by a wave of official-sector purchases on a scale not seen in decades. Central banks, led by emerging-market reserve managers diversifying away from dollar assets after the sanctions on Russian reserves, have been buying at a historic pace (Aizenman and Inoue, 2013; Arslanalp, Eichengreen and Simpson-Bell, 2023, 2025). As the identity of the marginal buyer has changed, a growing body of commentary, academic and market-based alike, argues that the "pricing power" of gold is migrating eastward, away from the traditional London–New York axis and toward Asian trading hours and Asian exchanges (Sobti, Sehgal and Ilango, 2021; Haupt, 2025). Silver, whose investment demand increasingly co-moves with gold, has swung even more violently over the same period. What is missing, and what this paper tries to supply, is direct evidence

on *when* within the 24-hour trading day the cumulative appreciation of the two metals is actually produced.

We ask three questions. How are the cumulative returns of gold and silver distributed across the Asian, European, and US trading sessions? Is that pattern stable across calendar years, and does it survive, or intensify, in the post-2022 bull market regime? And do session returns contain exploitable short-horizon predictability, meaning can a simple cross-session momentum rule survive transaction costs?

To answer them we use hourly spot prices of XAU/USD and XAG/USD from the Stooq global database, covering 22 July 2024 to 7 August 2026 (about 2.04 years, 529 trading days, 1,587 session observations per metal). The design follows the decomposition logic of Lou, Polk and Skouras (2019), who split equity returns into overnight and intraday pieces; we instead split the 24-hour precious-metals day into three UTC-defined sessions and chain the session returns in temporal order, so that session log returns add up exactly to the total log return. From there the computations are straightforward: each session's contribution to cumulative performance, its volatility and up-frequency, year by year, plus a backtest of a momentum rule that goes long (or flat) in session t+1 when session t's return was positive, run both over the full session cycle and restricted to individual sessions, net of a 0.02% cost per position change.

The headline result is that the Asian session is the dominant pricing session for both metals over the sample. It accounts for 77.2% of gold's cumulative log return and 61.7% of silver's, its contribution is positive in every calendar year, and it absorbed the sell-offs the US session produced, most dramatically in 2026 when the US-session contribution swung to −0.169 log points for gold and −0.423 for silver. The US session is a different animal. It carries the highest volatility of the three sessions alongside a zero-or-negative drift, a separation between information share and return contribution that echoes, at the session level, Hauptfleisch, Putniņš and Lucey (2016) on New York's dominance of information shares and Bondarenko and Muravyev (2023) on equity returns accruing at specific clock times rather than uniformly with volatility. Session returns also display economically meaningful but fragile momentum, concentrated in continuation into Asian hours; once realistic costs are charged, only a long-only, all-session implementation remains viable, at a net Sharpe of 1.68 in gold (versus 1.40 for buy-and-hold) with the maximum drawdown roughly halved, and 0.84 in silver with the drawdown improved from −53.6% to −26.2%.

In terms of related work, the closest antecedents sit in three places. Sobti, Sehgal and Ilango (2021) measure information shares across five global gold trading segments using futures and find a rising Shanghai share; we measure *return* contributions rather than information shares, use spot rather than futures prices, and add silver and a post-2022 sample. Lou, Polk and Skouras (2019) and the overnight-versus-intraday literature that followed (Bogousslavsky, 2021; Hendershott, Livdan and Rösch, 2020; Boyarchenko, Larsen and Whelan, 2023) work with a two-way split; we extend it to a three-way global session decomposition in a 24-hour commodity market where, unlike equities, there is no unique "close" to anchor on. Haupt (2025), a non-peer-reviewed decomposition of LBMA AM/PM fixes, is closest in spirit, and we add an academic design with a continuous hourly chain, formal statistics, and strategy backtests. We are not aware of any prior study that jointly decomposes gold *and* silver returns across three global sessions, links the decomposition to cross-session momentum backtests, and covers the post-2022 central-bank-buying regime. One caveat runs through everything below: our sample is short and coincides with a historic bull market, so the patterns we

document describe the current regime, not a time-invariant law, and we try to keep that in view at each step.

The question matters beyond its academic interest. Reserve managers deciding when to execute large official purchases, refiners and bullion banks managing inventory across time zones, and asset allocators timing entry into gold and silver exposures all face an intraday scheduling problem. If cumulative appreciation is systematically concentrated in particular hours, execution timing becomes a first-order determinant of realized cost. If session returns are also predictable from the immediately preceding session, then even a passive buy-and-hold exposure can be improved with a near-trivial timing rule. Our long-only backtest speaks to the second possibility, and we evaluate it with transaction costs that we believe are conservative for institutional execution in spot bullion or highly liquid futures.

Section 2 reviews the related literature along four lines. Section 3 describes the data and the design, Section 4 reports the decomposition, Section 5 the momentum backtests, and Section 6 discusses mechanisms. Section 7 lists the limitations, and Section 8 concludes.

## 2. Literature Review

### 2.1 Overnight versus intraday return decomposition

A large literature documents that average stock returns accrue disproportionately in one part of the trading day. Cooper, Cliff and Gulen (2008) show that the US equity risk premium over long horizons is earned entirely overnight, with intraday returns near zero or negative. Lou, Polk and Skouras (2019) frame this as a "tug of war" between overnight and intraday expected returns, driven by clienteles that persistently demand or supply at different times of day; Lou, Polk and Skouras (2024) take the framework international, and Lu, Zhang, Reinert and Cucuringu (2025) model the resulting lead-lag structure as a tug-of-war network. The proposed explanations are eclectic: sentiment (Aboody, Even-Tov, Lehavy and Trueman, 2018, who use overnight returns as a firm-level sentiment proxy), attention at the open (Berkman, Koch, Tuttle and Zhang, 2012), cross-sectional heterogeneity (Bogousslavsky, 2021; Branch and Ma, 2012). The pattern itself is not stable. Hendershott, Livdan and Rösch (2020) document that the "night" no longer dominates the "day" in recent decades, and Akbas, Boehmer, Erturk and Sorescu (2021) tie the overnight/intraday split to short-selling constraints. Boyarchenko, Larsen and Whelan (2023) document the "overnight drift" across asset classes, and Bondarenko and Muravyev (2023) push the clock-time logic furthest: the entire US equity market return can be traced to returns earned around the European open. Our three-way session decomposition of gold and silver is a direct analogue of this literature for a market that trades around the clock and therefore has no natural overnight/intraday split. The closest antecedent is Haupt (2025), a non-peer-reviewed analysis that splits LBMA AM and PM gold fixes into "eastern" and "western" components and reports a rising eastern share, which our design corroborates with continuous hourly data.

Two features of this literature matter for what follows. The two-way night/day split that is natural in equity markets has no obvious counterpart in a continuously traded commodity: with three major trading centres, the tug of war is potentially three-way, and its outcome need not resemble the equity pattern. And the equity findings are themselves regime-dependent, as the fading overnight premium in Hendershott, Livdan and Rösch (2020) shows, which is one reason we report the decomposition year by year rather than as a single full-sample table.

### 2.2 Gold price discovery across markets and time zones

A second literature asks *where* gold prices are set. Hauptfleisch, Putniņš and Lucey (2016), using information-share methods, find that New York (COMEX) contributes more to gold price discovery than London despite London's larger trading volumes. Lucey, Larkin and O'Connor (2013, 2014) show that pricing leadership among London, New York, and other centres is time-varying, with bidirectional volatility spillovers, and Xu and Fung (2005) document cross-market information flows between COMEX and TOCOM precious-metals futures. On Asia, Jin, Li, Wang and Yang (2018) examine price discovery in Chinese gold markets, and Sobti, Sehgal and Ilango (2021) partition the 24-hour day into five trading segments and estimate that the Shanghai Futures Exchange's information share rose to roughly 15–26%, evidence of rising Asian influence that our return-based results complement. Watkins and McAleer (2018) characterize the Tokyo session as liquidity-driven and the New York session as informed, anticipating the volatility-versus-drift asymmetry we find. Ivanov (2013) studies the impact of gold and silver ETFs on price discovery; Caminschi and Heaney (2014) analyze the London fixing mechanism; Narayan and Sharma (2018) relate gold returns to broader market conditions. None of these papers computes the *cumulative return contribution* of each session to the metals' total appreciation, which is the quantity of direct interest for the "eastward shift of pricing power" debate.

### 2.3 Intraday momentum and return predictability

Gao, Han, Li and Zhou (2018) establish market intraday momentum: the first half-hour return predicts the last half-hour return in equity index markets, a pattern Zhang, Ma, Shen and Zhang (2019) show holds internationally. Time-series momentum at longer horizons is documented by Moskowitz, Ooi and Pedersen (2012). At the session level, Lu, Malliaris and Qin (2023) show that the overnight-minus-intraday return spread predicts future returns. In commodity markets, the Journal of Commodity Markets (2025) reports a "night effect" in Chinese gold and silver futures, whereby the first half-hour of the night session dominates prediction, and a study of commodity ETFs (PMC7480318, 2020) documents half-hour-level predictability in GLD and SLV. Our momentum design, the sign of session t's return predicting session t+1's return, is a coarse but transparent member of this family. What it adds is a time-zone dimension: predictability concentrates in continuation *into the Asian session*, something single-market studies cannot see.

### 2.4 Central bank gold demand, safe-haven properties, and the post-2022 regime

Baur and Lucey (2010) define the safe-haven and hedge properties of gold; Baur and McDermott (2010) extend the analysis internationally; Baur and Smales (2020) show gold hedges geopolitical risk; Lucey and Li (2015) examine which precious metals serve as safe havens and find silver's role weaker and time-varying; O'Connor, Lucey, Batten and Baur (2015) survey the financial economics of gold. On the demand side, Aizenman and Inoue (2013) document the determinants of central-bank gold holding, and Arslanalp, Eichengreen and Simpson-Bell (2023, 2025) link the post-2022 surge in official purchases to reserve diversification and sanctions risk. That regime change is the economic backdrop of our sample, and a candidate explanation for why Asian-hours buying pressure may have become dominant, though our design cannot identify the buyers directly.

### 2.5 Research gap

What is missing from this literature is not hard to state. The decomposition work in Section 2.1 is equity-centric and two-way (night versus day), while the price-discovery work in Section 2.2 reports information shares rather than cumulative return contributions. The momentum work in Section 2.3 rarely connects predictability to global time zones in 24-hour commodity markets. And almost no academic evidence covers the post-2022 central-bank-buying regime. This paper takes up all three at once, for gold and silver, within a single consistent dataset and design.

## 3. Data and Methodology

### 3.1 Data

We use hourly bars of spot gold (XAU/USD) and spot silver (XAG/USD) from the Stooq global hourly historical database (the `h_world_txt` package, stooq.com), retrieved on 10 August 2026. The sample runs from 22 July 2024 01:00 UTC to 7 August 2026 23:00 UTC, comprising 12,138 hourly bars for gold and 12,126 for silver, approximately 2.04 years and 529 trading days, hence 1,587 session observations per metal.

A word on how we ended up with this source, since it shaped the paper. The original plan called for COMEX futures (GC/SI) hourly data pulled from Yahoo Finance, but the API rate-limited our research environment (repeated HTTP 429 responses), and the alternative hourly sources we tried, Dukascopy and HistData, were not reachable from it either. Rather than hold the project hostage to one data vendor, we switched to Stooq spot data. Stooq's hourly history only reaches back about two years, so the sample is confined to 2024H2–2026H1, which happens to coincide with a historic bull market in both metals. Section 7 returns to what that restriction does and does not imply.

Timestamp validation confirms the bars are stamped in UTC: gold volatility peaks in the 15:00–17:00 UTC bars, coincident with COMEX floor hours and US macroeconomic releases, and is lowest in the 23:00 and 06:00 UTC bars. There remains a ±1 hour ambiguity in how the hourly bars are labeled (open-time versus close-time stamping), but with session boundaries 6.5–8 hours wide, that ambiguity cannot change the relative ordering of the three sessions.

### 3.2 Session definitions

We partition the 24-hour UTC day into three sessions, assigning each hourly bar by its opening time as shown in Table 1.

**Table 1. Session definitions (UTC, by bar open time)**

| Session | UTC window | Hourly bars included |
|---|---|---|
| Asia | 00:00–08:00 | bars 00–07 |
| Europe | 08:00–14:30 | bars 08–14 |
| US + late | 14:30–24:00 | bars 15–23 (incl. 21–24 tail) |

The US session and the 21:00–24:00 late window are merged into a single "US + late" session spanning 14:30–24:00 UTC. Sessions are chained in temporal order, previous-day US, then Asia, then

Europe, then US, so that the Sunday 22:00–24:00 opening bars naturally precede Monday's Asian session. The session return is the last hourly close within the session divided by the last hourly close of the preceding session, minus one; weekends and holidays are bridged automatically by the chain. Because the chain is exact, session log returns sum over the sample to the total log return, which is what makes an additive decomposition of cumulative performance possible.

The windows deserve a little justification. Asia's 00:00–08:00 UTC corresponds to 08:00–16:00 Beijing time and 09:00–17:00 Tokyo time, covering the core of Shanghai Gold Exchange and Shanghai Futures Exchange gold and silver trading as well as TOCOM hours. The European window, 08:00–14:30 UTC, covers London's morning and afternoon activity up to, but excluding, the US macroeconomic releases that cluster at 12:30–13:30 UTC and the COMEX floor open. The 14:30 UTC boundary (9:30 US Eastern) approximates the New York equity open and the heart of COMEX trading. The 21:00–24:00 tail, thin Globex and OTC activity after the COMEX settlement, is merged with the US session because preliminary inspection showed it behaving statistically like the late US afternoon rather than like Asia. These are approximations. Daylight-saving transitions shift local session times by one hour relative to UTC, and we treat that shift as acceptable given the width of the windows.

### 3.3 Momentum strategy definition

The trading signal is the sign of the session return: if session t's return is positive, the strategy is long in session t+1; if negative, it is short (long-short version) or flat (long-only version). Variants apply the rule (i) across all sessions, or (ii) holding only during the Asian, only during the European, or only during the US session (flat otherwise). Transaction costs of 0.02% are charged on every position change, including entries, exits, and reversals; a long-short reversal therefore costs two changes.

### 3.4 Performance statistics

For each strategy we report annualized geometric return, annualized volatility (scaled by √N at the actual session frequency), the Sharpe ratio with a zero risk-free rate, maximum drawdown, and the win rate (the fraction of held sessions with positive returns). For the decomposition we report per-session mean returns and standard deviations in basis points, cumulative log returns, contribution shares (session cumulative log return divided by the total log return; the three shares sum to 100%), and the probability of a positive session return. Autocorrelations are computed on the session-return chain overall and conditional on the destination session.

## 4. Empirical Results

### 4.1 Session return statistics

Over the full sample, gold's cumulative log return is +0.591 (approximately +81%) and silver's is +0.782 (approximately +119%). Tables 2 and 3 decompose these totals by session.

**Table 2. Gold (XAU/USD) session return statistics, 2024-07-22 to 2026-08-07**

| Session | Mean return (bp) | Std. dev. (bp) | Cumulative log return | Contribution share | Prob. of up session |
|---|---|---|---|---|---|

| | | | | | |
|---|---|---|---|---|---|
| Asia 00–08 | +8.94 | 79.3 | +0.456 | **+77.2%** | 56.7% |
| Europe 08–14:30 | +4.30 | 70.2 | +0.214 | +36.2% | 55.2% |
| US + late 14:30–24 | −1.09 | 91.0 | −0.079 | **−13.4%** | 52.9% |

**Table 3. Silver (XAG/USD) session return statistics, 2024-07-22 to 2026-08-07**

| Session | Mean return (bp) | Std. dev. (bp) | Cumulative log return | Contribution share | Prob. of up session |
|---|---|---|---|---|---|
| Asia 00–08 | +10.35 | 156.9 | +0.482 | **+61.7%** | 53.1% |
| Europe 08–14:30 | +7.92 | 143.7 | +0.364 | +46.6% | 55.0% |
| US + late 14:30–24 | +0.61 | 189.8 | −0.065 | **−8.2%** | 51.0% |

*Note:* Contribution share = session cumulative log return / full-sample log return; shares sum to 100%.

The pattern that jumps out of the tables is the asymmetry between Asia and the US. The Asian session is the largest positive contributor for both metals, with 77.2% of gold's and 61.7% of silver's cumulative gain accruing between 00:00 and 08:00 UTC. Europe adds a stable positive share (36.2% for gold, 46.6% for silver); Asia plus Europe together exceed 100% of the total, which is arithmetic rather than magic, since the third contribution is negative. And that third contribution is the interesting one. The US session, including the 21:00–24:00 tail, posts the *highest* volatility of the three sessions, 91.0 bp for gold and 189.8 bp for silver against 79.3 and 156.9 bp in Asia, yet a *negative* cumulative contribution (−13.4% and −8.2%). The probability of an up session is similar across sessions (51–57%), so the asymmetry lives in average magnitudes rather than frequencies. Asia buys; the US session churns and gives back. Volatility and drift are decoupled across the clock.

### 4.2 Cumulative contribution

Figure 1 plots the cumulative net asset value of each session's return stream for gold alongside buy-and-hold; Figure 2 does the same for silver.

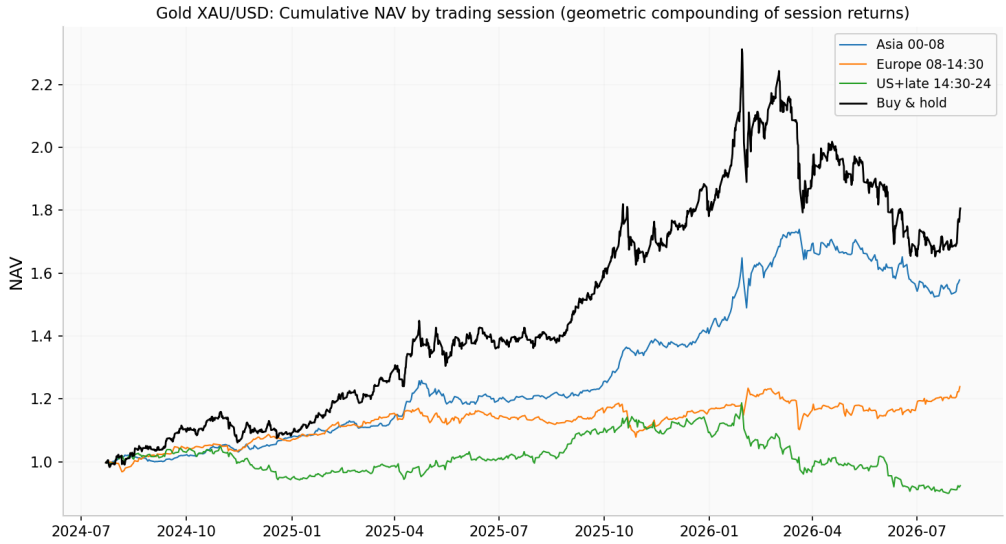



**Figure 1.** *Gold: cumulative NAV by session.*

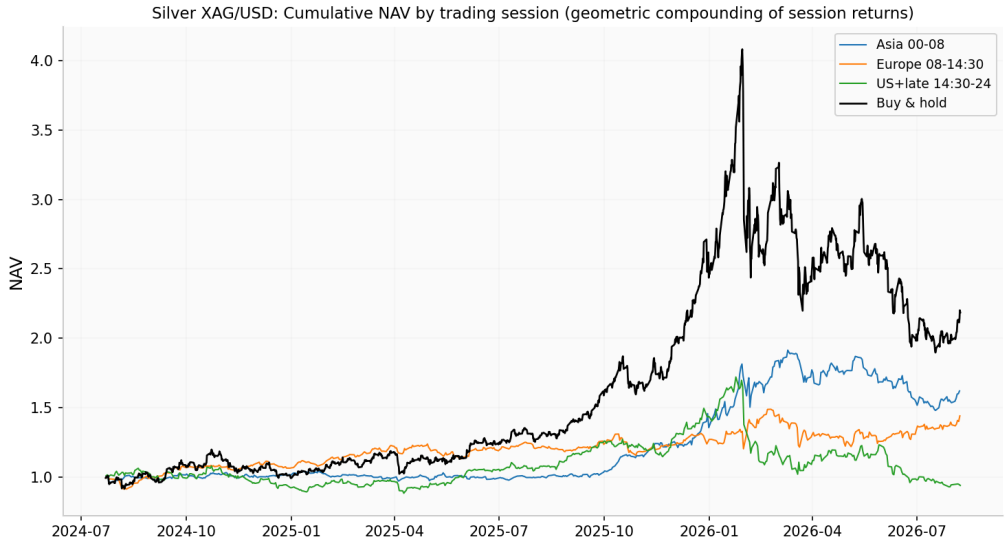



**Figure 2.** *Silver: cumulative NAV by session.*

In both figures the Asian-session line nearly traces every upward leg of the buy-and-hold path, while the US-session line drifts sideways-to-downward over most of the sample. The picture matches the tables: over this sample, the marginal pricing of gold and silver happened disproportionately during Asian hours.

### 4.3 Year-by-year decomposition

Tables 4 and 5 report the decomposition by calendar year (log returns), and Figures 3 and 4 display them.

**Table 4. Gold: yearly session contributions (log returns)**

| Year | Asia | Europe | US + late |
|---|---|---|---|
| 2024 (H2) | +0.074 | +0.067 | −0.054 |

| | | | |
|---|---|---|---|
| 2025 | +0.268 | +0.087 | +0.144 |
| 2026 (to Aug.) | +0.114 | +0.061 | −0.169 |

**Table 5. Silver: yearly session contributions (log returns)**

| Year | Asia | Europe | US + late |
|---|---|---|---|
| 2024 (H2) | +0.011 | +0.063 | −0.081 |
| 2025 | +0.292 | +0.176 | +0.440 |
| 2026 (to Aug.) | +0.179 | +0.126 | −0.423 |

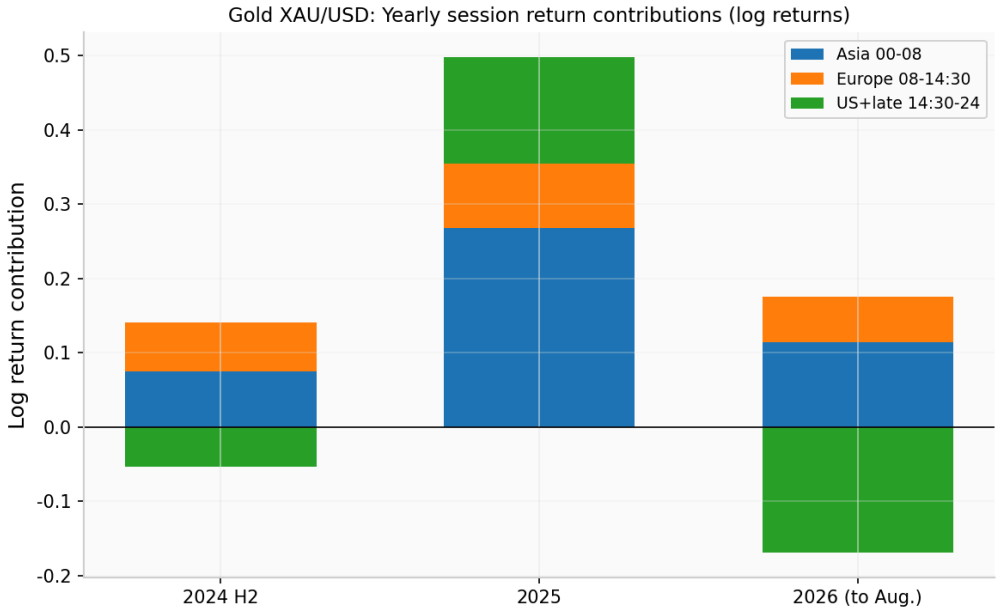



**Figure 3.** *Gold: yearly session contributions.*

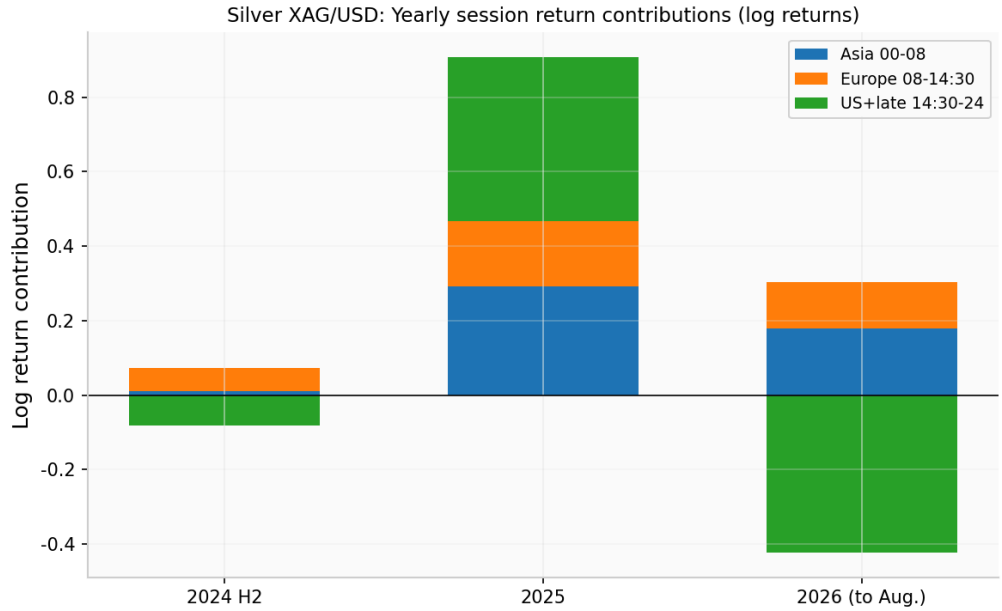



**Figure 4.** *Silver: yearly session contributions.*

Several things here are worth pausing on. The Asian contribution is positive in *every* calendar year and largest in 2025, so the Asian dominance is not an artifact of one episode. During the 2025 acceleration of the bull market the US session contributed positively too, most visibly in silver (+0.440), which tells us that in trending melt-up phases the buying pressure spills over into US hours. Then came 2026. The correction was produced almost entirely during the US session: its contribution swung to −0.169 for gold and a remarkable −0.423 for silver, while Asia kept accumulating (+0.114 and +0.179). In share terms, Asia accounted for roughly 85% of gold's gain in 2024 H2, and in 2026 year-to-date Asia plus Europe supplied all of the positive pricing while the US session ran negative. The claim of an eastward shift in pricing power is thus supported within this sample not only in levels but in the time series of contributions as well.

### 4.4 Autocorrelation structure

The lag-1 autocorrelation of session returns is modest overall, +0.02 for gold (same-sign proportion 50.0%) and +0.05 for silver (47.8%), but sharply concentrated: continuation *into the Asian session* carries an autocorrelation of +0.13 for gold and +0.12 for silver, while autocorrelations into the European and US sessions sit near zero or below. This is where the strategy results of the next section come from. Whatever short-horizon momentum exists in these markets is momentum into Asian hours.

## 5. Trading Strategy Backtests

Tables 6 and 7 report backtest results for gold and silver, respectively, with a transaction cost of 0.02% per position change applied to the net variants.

**Table 6. Gold (XAU/USD) momentum backtests (cost 0.02% per change)**

| Strategy | Ann. return | Ann. vol. | Sharpe | Max drawdown | Win rate |
|---|---|---|---|---|---|
| Buy-and-hold | 33.6% | 22.5% | 1.40 | −28.5% | 54.9% |

| | | | | | |
|---|---|---|---|---|---|
| All sessions, long-short (gross) | 37.1% | 22.5% | 1.51 | −11.8% | 50.1% |
| All sessions, long-short (net) | 17.4% | 22.5% | 0.82 | −15.0% | 48.6% |
| All sessions, long-only (gross) | 37.4% | 14.9% | 2.20 | −14.0% | 54.6% |
| All sessions, long-only (net) | **27.2%** | 15.0% | **1.68** | −15.7% | 54.3% |
| Asia only, long-short (net) | −0.5% | 12.8% | 0.02 | −19.8% | 45.3% |
| Asia only, long-only (net) | 11.8% | 8.4% | 1.37 | −9.3% | 52.3% |
| Europe only, long-short (net) | −0.9% | 11.3% | −0.02 | −13.1% | 52.1% |
| Europe only, long-only (net) | 4.7% | 6.8% | 0.71 | −5.3% | 55.3% |
| US only, long-short (net) | 1.9% | 14.6% | 0.21 | −15.8% | 49.7% |
| US only, long-only (net) | −1.0% | 10.3% | −0.04 | −12.7% | 52.7% |

**Table 7. Silver (XAG/USD) momentum backtests (cost 0.02% per change)**

| Strategy | Ann. return | Ann. vol. | Sharpe | Max drawdown | Win rate |
|---|---|---|---|---|---|
| Buy-and-hold | 46.6% | 45.9% | 1.07 | −53.6% | 53.1% |
| All sessions, long-short (gross) | 3.9% | 45.9% | 0.31 | −43.5% | 47.9% |
| All sessions, long-short (net) | −11.6% | 45.9% | −0.04 | −50.2% | 46.9% |
| All sessions, long-only (gross) | 32.4% | 28.7% | 1.12 | −24.5% | 51.1% |
| All sessions, long-only (net) | **22.1%** | 28.7% | **0.84** | −26.2% | 51.0% |
| Asia only, long-short (net) | −5.3% | 25.3% | −0.09 | −41.9% | 45.3% |
| Asia only, long-only (net) | 11.8% | 15.5% | 0.79 | −15.2% | 48.9% |
| Europe only, long-short (net) | −0.7% | 23.1% | 0.09 | −23.2% | 49.2% |
| Europe only, long-only (net) | 10.7% | 14.0% | 0.79 | −11.1% | 54.1% |
| US only, long-short (net) | −19.0% | 30.5% | −0.54 | −47.0% | 47.5% |
| US only, long-only (net) | −9.3% | 19.6% | −0.40 | −37.0% | 48.8% |

A few conclusions stand out.

Before costs, the momentum signal works. In gold, the all-session long-short variant earns a gross Sharpe of 1.51 and the long-only variant 2.20, both above buy-and-hold's 1.40, with the drawdown roughly halved (−11.8% and −14.0% versus −28.5%). The long-only version cuts volatility from 22.5% to 14.9% simply by standing aside after negative sessions.

Transaction costs are where most of the variants die. The all-session long-short rule changes position roughly 780 times per year (times two sides on reversals), a drag of about 15–20 percentage points annually: gold's net Sharpe falls from 1.51 to 0.82, and silver's turns negative (−0.04). The one variant that survives is the all-session long-only rule. In gold it delivers a net Sharpe of 1.68 with a maximum drawdown of −15.7% (versus −28.5% for buy-and-hold); in silver, a net Sharpe of 0.84 with the drawdown improved from −53.6% to −26.2%. Put plainly, "ride the trend, stand aside after down

sessions" beats buy-and-hold on a risk-adjusted basis, at the cost of some absolute return (27.2% versus 33.6% in gold; 22.1% versus 46.6% in silver).

The signal is concentrated in Asian and European hours. Gross long-only Sharpe ratios by session (gold: Asia 2.01, Europe 1.57; silver: Asia 1.13, Europe 1.19) mirror the autocorrelation structure of Section 4.4. Every momentum variant restricted to the US session loses money net of costs; silver's US-only long-short variant gives up 19.0% annualized at a Sharpe of −0.54. That is what one would expect if US hours combine high variance with no drift. If anything, the US session looks better suited to contrarian than to trend-following rules, though we do not push in that direction here.

Gold and silver differ in degree rather than in kind. The session structure is the same in both, Asian pricing dominance, US-session give-back, viable long-only momentum, but silver runs at roughly twice gold's volatility (45.9% versus 22.5% annualized for buy-and-hold), its signal-to-noise ratio is lower, and every strategy's Sharpe sits below its gold counterpart. Because silver's buy-and-hold drawdown is extreme (−53.6%), the risk-management value of the long-only overlay is correspondingly larger in silver.

## 6. Discussion

### 6.1 Mechanisms: why the Asian session?

Our design is descriptive and cannot identify traders, but the pattern fits the demand structure of the post-2022 regime. Official-sector accumulation is concentrated among Asian and emerging-market central banks (Arslanalp, Eichengreen and Simpson-Bell, 2023, 2025; Aizenman and Inoue, 2013), whose execution naturally falls in Asian business hours; persistent price-insensitive buying of that kind would produce exactly what we observe, a positive Asian-hours drift that nobody arbitrages away intraday. The Shanghai premium and the rising information share of Chinese exchanges (Sobti, Sehgal and Ilango, 2021; Jin, Li, Wang and Yang, 2018) point the same way: physical and futures demand originating in China has become marginal for price formation. It is also telling that Asian-session returns are the most *persistent* (Section 4.4). Slow-moving, flow-based demand behaves that way; news-based trading does not. This is the clientele interpretation of Lou, Polk and Skouras (2019), transplanted to a 24-hour commodity market.

The economic magnitude deserves attention. A mean Asian-session return of +8.94 bp in gold against an up-probability of only 56.7% implies a strongly right-skewed session distribution: the average up Asian session has to be considerably larger than the average down session. The US session shows the mirror image, a near-coin-flip up frequency (52.9%) combined with a *negative* mean, which implies reverse skew, occasional large down-moves that dominate the cumulative contribution. The 2026 subsample, in which the US session lost 0.169 log points in gold while Asia gained 0.114, is the extreme expression of this asymmetry. One way to read the three sessions is as three distinct return-generating processes running in sequence each day: persistent accumulation in Asia, transition in Europe, high-variance repricing in US hours.

### 6.2 The US session: information share without return contribution

The US session combines the highest volatility with a negative cumulative drift, and that divorces two notions of "where prices are made." On information share, New York has been found dominant (Hauptfleisch, Putniņš and Lucey, 2016), and US macroeconomic announcements clearly move gold

violently; our volatility numbers confirm as much. But processing information is not the same as producing cumulative return. A session can incorporate news in both directions, and host profit-taking after Asian-hours advances, while contributing nothing to the drift. The finding parallels Bondarenko and Muravyev (2023), who show that US equity returns accrue at a specific clock time rather than where volatility is highest, and Watkins and McAleer (2018), who characterize Tokyo-hours trading as liquidity-driven and New York-hours trading as informed. Our results suggest that, in the current regime, the *informed but mean-reverting* US session and the *flow-driven, drifting* Asian session play complementary roles. The 2026 experience, when nearly the entire correction occurred in US hours while Asia accumulated, fits the same story: fast money sells on macro news in US hours; patient demand absorbs the supply in Asian hours.

### 6.3 Gold versus silver

Silver's session structure mirrors gold's, at roughly double the volatility and a weaker signal-to-noise ratio. The Asian contribution share is smaller for silver (61.7% versus 77.2%), mainly because silver's 2025 US-session contribution was strongly positive (+0.440) and its 2026 US-session give-back correspondingly deeper (−0.423): silver amplifies whatever the US session does. That fits silver's dual identity as monetary and industrial metal, and its thinner liquidity. For practitioners, the contrast means session timing matters more, and pure momentum less, in silver than in gold.

### 6.4 Implications for market participants

The practical implications differ by agent. For official-sector accumulators, upward drift during Asian hours argues for patient execution *within* the Asian session, spreading purchases across hours rather than concentrating them at liquid fixes, as a way to soften the price impact of a flow that is itself plausibly part of the drift we measure. We recognize the endogeneity in that suggestion and offer it as a hypothesis, not a prescription. For Western-based allocators accessing gold through London or New York hours, a systematically Asian-hours-priced asset embeds an execution-timing cost: entering during US hours means, on average over this sample, buying after the day's appreciation has largely happened. For risk managers, the drawdown profile is as informative as the drift. The largest losses, especially in silver, are generated in US hours, so intraday risk limits and margin policies calibrated to the US session's 91–190 bp standard deviation (versus 79–157 bp in Asia) are warranted. And for investors unwilling to time sessions at all, the long-only overlay, net Sharpe 1.68 in gold with the maximum drawdown cut from −28.5% to −15.7%, suggests that a coarse "stand aside after a down session" rule already captures much of the benefit, though the fragility of the long-short variant to costs warns against more aggressive versions.

### 6.5 Relation to the eastward-shift debate

Our results provide return-based support for the view that gold's marginal pricing has shifted toward Asian hours, complementing the information-share evidence of Sobti, Sehgal and Ilango (2021) and the fix-based, non-academic decomposition of Haupt (2025). Two cautions apply. Return contribution is not price discovery in the microstructure sense; a high-contribution session need not be the most informationally efficient one. And our sample coincides with an exceptional bull market and with the peak of the central-bank buying wave, so whether Asian dominance persists through a sideways or bear market is an open question our data cannot answer.

The backtest evidence deserves a caveat of its own, on statistical power. A lag-1 autocorrelation of +0.13 into the Asian session, estimated on roughly 529 observations, is only marginally significant by conventional standards, and we have not adjusted for the multiplicity of variants in Tables 6 and 7. We read the strategy results as *suggestive and economically coherent*: the gross Sharpe ordering, the cost sensitivity, and the session concentration all line up with the autocorrelation structure and with the decomposition. We would not read them as ironclad evidence of a standalone anomaly.

## 7. Limitations

The findings come with qualifications, and they are not minor. The sample spans only about two years (2024 H2–2026 H1) and lies entirely inside a historic, one-directional bull market; the high Asian-session contribution and the bull-market structure may reinforce each other, and the patterns need re-examination in range-bound or bear regimes. We would have liked a longer hourly history, but none was available from the sources we could reach, and daily data, while longer, cannot support an intraday session decomposition at all. We use spot XAU/USD and XAG/USD rather than COMEX futures; spot–futures basis differences exist, though the session *structure* should be common to both. The ±1-hour labeling ambiguity in the hourly bar timestamps (open- versus close-time stamping) does not affect the relative conclusions across three wide sessions but could shift marginal statistics. The backtests assume execution at session boundaries with a flat 0.02% cost and no market impact; in fast markets, realistic execution could be costlier. Finally, all strategy performance is evaluated in-sample over a favorable regime. Out-of-sample validation on future data is essential before anyone trades on this.

## 8. Conclusion

Over July 2024 to August 2026, the appreciation of gold and silver was produced overwhelmingly outside US trading hours. The Asian session accounts for 77.2% of gold's and 61.7% of silver's cumulative log return, contributes positively in every calendar year, and kept accumulating through the 2026 correction that the US session almost single-handedly produced. The US session remains the most volatile of the three but subtracted from cumulative returns, separating information processing from return generation. Session returns embed a short-horizon momentum concentrated in continuation into Asian hours; after realistic costs, only a long-only, all-session implementation survives, roughly halving drawdowns relative to buy-and-hold (net Sharpe 1.68 in gold; 0.84 in silver). The evidence is consistent with an eastward migration of marginal pricing power in the post-2022 regime of central-bank accumulation. The sample is short and regime-specific, and we would caution against extrapolating it too far.

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